

Tag	Description
<code><video></code>	Defines a video or movie
<code><source></code>	Defines multiple media resources for media elements, such as <code><video></code> and <code><audio></code>
<code><track></code>	Defines text tracks in media players

HTML Text Formatting

HTML CONTAINS SEVERAL ELEMENTS FOR DEFINING TEXT WITH A SPECIAL MEANING.

Example

This text is bold

This text is italic

This is _{subscript} and ^{superscript}

HTML FORMATTING ELEMENTS

Formatting elements were designed to display special types of text:

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

HTML AND ELEMENTS

The HTML element defines bold text, without any extra importance.

The HTML element defines text with strong importance. The content inside is typically displayed in bold.

HTML <I> AND ELEMENTS

The HTML <i> element defines a part of text in an alternate voice or mood. The content inside is typically displayed in italic.

Tip: The <i> tag is often used to indicate a technical term, a phrase from another language, a thought, a ship name, etc.

Example

```
<i>This text is italic</i>
```

The HTML element defines emphasized text. The content inside is typically displayed in italic.

Tip: A screen reader will pronounce the words in with an emphasis, using verbal stress.

Example

```
<em>This text is emphasized</em>
```

HTML <SMALL> ELEMENT

The HTML <small> element defines smaller text:

Example

```
<small>This is some smaller text.</small>
```

HTML <MARK> ELEMENT

The HTML <mark> element defines text that should be marked or highlighted:

Example

<p>Do not forget to buy <mark>milk</mark> today.</p>

HTML ELEMENT

The HTML element defines text that has been deleted from a document. Browsers will usually strike a line through deleted text:

Example

<p>My favorite color is blue red.</p>

HTML <INS> ELEMENT

The HTML <ins> element defines a text that has been inserted into a document. Browsers will usually underline inserted text:

Example

<p>My favorite color is blue <ins>red</ins>.</p>

HTML <SUB> ELEMENT

The HTML <sub> element defines subscript text. Subscript text appears half a character below the normal line, and is sometimes rendered in a smaller font. Subscript text can be used for chemical formulas, like H₂O:

Example

<p>This is _{subscripted} text.</p>

HTML <SUP> ELEMENT

The HTML <sup> element defines superscript text. Superscript text appears half a character above

the normal line, and is sometimes rendered in a smaller font. Superscript text can be used for footnotes, like WWW:

Example

<p>This is ^{superscripted} text.</p>

HTML Multimedia

MULTIMEDIA ON THE WEB IS SOUND, MUSIC, VIDEOS, MOVIES, AND ANIMATIONS.

WHAT IS MULTIMEDIA?

Multimedia comes in many different formats. It can be almost anything you can hear or see, like images, music, sound, videos, records, films, animations, and more.

Web pages often contain multimedia elements of different types and formats.

BROWSER SUPPORT

The first web browsers had support for text only, limited to a single font in a single color.

Later came browsers with support for colors, fonts, images, and multimedia!

MULTIMEDIA FORMATS

Multimedia elements (like audio or video) are stored in media files.

The most common way to discover the type of a file, is to look at the file extension.

Multimedia files have formats and different extensions like: .wav, .mp3, .mp4, .mpg, .wmv, and .avi.

COMMON VIDEO FORMATS



There are many video formats out there.

The MP4, WebM, and Ogg formats are supported by HTML.

The MP4 format is recommended by YouTube.

Note: Only MP4, WebM, and Ogg video are supported by the HTML standard.

THE HTML <VIDEO> ELEMENT

To show a video in HTML, use the <video> element:

Example

```
<video width="320" height="240" controls>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
Your browser does not support the video tag.
</video>
```

HOW IT WORKS

- The controls attribute adds video controls, like play, pause, and volume.
- It is a good idea to always include width and height attributes. If height and width are not set, the page might flicker while the video loads.
- The <source> element allows you to specify alternative video files which the browser may choose from. The browser will use the first recognized format.
- The text between the <video> and </video> tags will only be displayed in browsers that do not support the <video> element.

HTML <VIDEO> AUTOPLAY

To start a video automatically, use the autoplay attribute:

Example

```
<video width="320" height="240" autoplay>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
Your browser does not support the video tag.
</video>
```

Note: Chromium browsers do not allow autoplay in most cases. However, muted autoplay is always allowed.

Add muted after autoplay to let your video start playing automatically (but muted):

Example

```
<video width="320" height="240" autoplay muted>  
  <source src="movie.mp4" type="video/mp4">  
  <SOURCE src="movie.ogg" type="video/ogg">  
Your browser does not support the video tag.  
</video>
```

BROWSER SUPPORT

The numbers in the table specify the first browser version that fully supports the <video> element.

HTML Video Formats

There are three supported video formats: MP4, WebM, and Ogg. The browser support for the different formats is:

HTML VIDEO - MEDIA TYPES

File format	Media Type
mp4	Video/mp4
webM	Video/webm
Ogg	Video/ogg

HTML VIDEO - METHODS, PROPERTIES, AND EVENTS

The HTML DOM defines methods, properties, and events for the <video> element.

This allows you to load, play, and pause videos, as well as setting duration and volume.

There are also DOM events that can notify you when a video begins to play, is paused, etc.

HTML AUDIO

The HTML <audio> element is used to play an audio file on a web page.

THE HTML <AUDIO> ELEMENT

To play an audio file in HTML, use the <audio> element:

Example

```
<audio controls>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
Your browser does not support the audio element.
</audio>
```

HTML AUDIO - HOW IT WORKS

The controls attribute adds audio controls, like play, pause, and volume.

The <source> element allows you to specify alternative audio files which the browser may choose from. The browser will use the first recognized format.

The text between the <audio> and </audio> tags will only be displayed in browsers that do not support the <audio> element.

HTML <AUDIO> AUTOPLAY

To start an audio file automatically, use the autoplay attribute:

Example

```
<audio controls autoplay>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
Your browser does not support the audio element.
</audio>
```

Note: Chromium browsers do not allow autoplay in most cases. However, muted autoplay is always allowed.

Add muted after autoplay to let your audio file start playing automatically (but muted):

Example

```
<audio controls autoplay muted>  
  <source src="horse.ogg" type="audio/ogg">  
  <source src="horse.mp3" type="audio/mpeg">  
Your browser does not support the audio element.  
</audio>
```

HTML AUDIO FORMATS

There are three supported audio formats: MP3, WAV, and OGG. The browser support for the different formats is:

HTML AUDIO - MEDIA TYPES

File format	Media Type
Mp3	audio/mpeg
wav	audio/wav
Ogg	audio/ogg

HTML <IFRAME> TAG

Example

An inline frame is marked up as follows:

```
<iframe src="https://www.w3schools.com" title="W3Schools Free Online Web  
Tutorials"></iframe>
```

DEFINITION AND USAGE

The <iframe> tag specifies an inline frame.

An inline frame is used to embed another document within the current HTML document.

HTML FORMS

An HTML form is used to collect user input. The user input is most often sent to a server for processing.

- The `<form>` Element
 - The HTML `<form>` element is used to create an HTML form for user input:
`<form>`
 - *form elements*
 - `</form>`
 - The `<form>` element is a container for different types of input elements, such as: text fields, checkboxes, radio buttons, submit buttons, etc.
 - All the different form elements are covered in this chapter: [HTML Form Elements](#).
-

THE `<INPUT>` ELEMENT

The HTML `<input>` element is the most used form element.

An `<input>` element can be displayed in many ways, depending on the type attribute.

Here are some examples:

Type	Description
<code><input type="text"></code>	Displays a single-line text input field
<code><input type="radio"></code>	Displays a radio button (for selecting one of many choices)
<code><input type="checkbox"></code>	Displays a checkbox (for selecting zero or more of many choices)
<code><input type="submit"></code>	Displays a submit button (for submitting the form)
<code><input type="button"></code>	Displays a clickable button

The `<input type="text">` defines a single-line input field for text input.

Example

A form with input fields for text:

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname">
</form>
```

Note: The form itself is not visible. Also note that the default width of an input field is 20 characters.

THE `<LABEL>` ELEMENT

Notice the use of the `<label>` element in the example above.

The `<label>` tag defines a label for many form elements.

The `<label>` element is useful for screen-reader users, because the screen-reader will read out loud the label when the user focuses on the input element.

The `<label>` element also helps users who have difficulty clicking on very small regions (such as radio buttons or checkboxes) - because when the user clicks the text within the `<label>` element, it toggles the radio button/checkbox.

The `for` attribute of the `<label>` tag should be equal to the `id` attribute of the `<input>` element to bind them together.

RADIO BUTTONS

The `<input type="radio">` defines a radio button.

Radio buttons let a user select ONE of a limited number of choices.

Example

A form with radio buttons:

```
<p>Choose your favorite Web language:</p>

<form>
  <input type="radio" id="html" name="fav_language" value="HTML">
```

```
<label for="html">HTML</label><br>
<input type="radio" id="css" name="fav_language" value="CSS">
<label for="css">CSS</label><br>
<input type="radio" id="javascript" name="fav_language" value="JavaScript">
<label for="javascript">JavaScript</label>
</form>
```

CHECKBOXES

The `<input type="checkbox">` defines a **checkbox**.

Checkboxes let a user select ZERO or MORE options of a limited number of choices.

Example

A form with checkboxes:

```
<form>
<input type="checkbox" id="vehicle1" name="vehicle1" value="Bike">
<label for="vehicle1"> I have a bike</label><br>
<input type="checkbox" id="vehicle2" name="vehicle2" value="Car">
<label for="vehicle2"> I have a car</label><br>
<input type="checkbox" id="vehicle3" name="vehicle3" value="Boat">
<label for="vehicle3"> I have a boat</label>
</form>
```

THE SUBMIT BUTTON

The `<input type="submit">` defines a button for submitting the form data to a form-handler.

The form-handler is typically a file on the server with a script for processing input data.

The form-handler is specified in the form's action attribute.

Example

A form with a submit button:

```
<form action="/action_page.php">
<label for="fname">First name:</label><br>
<input type="text" id="fname" name="fname" value="John"><br>
<label for="lname">Last name:</label><br>
<input type="text" id="lname" name="lname" value="Doe"><br><br>
```

```
<input type="submit" value="Submit">
</form>
```

Example

This example will not submit the value of the "First name" input field:

```
<form action="/action_page.php">
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" value="John"><br><br>
  <input type="submit" value="Submit">
</form>
```

HTML SEMANTIC ELEMENTS

Semantic elements = elements with a meaning.

What are Semantic Elements?

A semantic element clearly describes its meaning to both the browser and the developer.

Examples of **non-semantic** elements: <div> and - Tells nothing about its content.

Examples of **semantic** elements: , <table>, and <article> - Clearly defines its content.

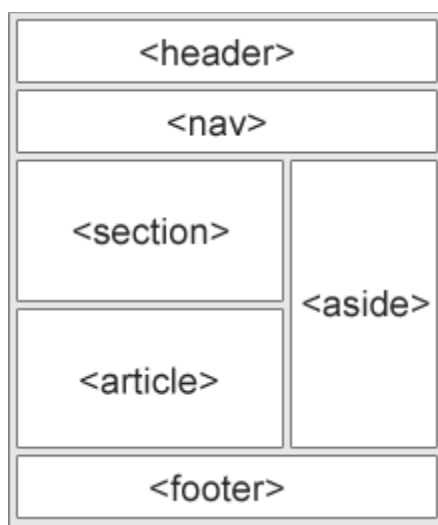
SEMANTIC ELEMENTS IN HTML

Many web sites contain HTML code like: <div id="nav"> <div class="header"> <div id="footer"> to indicate navigation, header, and footer.

In HTML there are several semantic elements that can be used to define different parts of a web page:

- <article>
- <aside>
- <details>
- <figcaption>
- <figure>

- `<footer>`
- `<header>`
- `<main>`
- `<mark>`
- `<nav>`
- `<section>`
- `<summary>`
- `<time>`



HTML `<section>` Element

The `<section>` element defines a section in a document.

According to W3C's HTML documentation: "A section is a thematic grouping of content, typically with a heading."

Examples of where a `<section>` element can be used:

- Chapters
- Introduction
- News items
- Contact information

A web page could normally be split into sections for introduction, content, and contact information.

Example

Two sections in a document:

```
<section>
```

```
<h1>WWF</h1>
```

```
<p>The World Wide Fund for Nature (WWF) is an international organization working on  
issues regarding the conservation, research and restoration of the environment, formerly  
named the World Wildlife Fund. WWF was founded in 1961.</p>
```

```
</section>
```

```
<section>
```

```
<h1>WWF's Panda symbol</h1>
```

```
<p>The Panda has become the symbol of WWF. The well-known panda logo of WWF  
originated from a panda named Chi Chi that was transferred from the Beijing Zoo to the  
London Zoo in the same year of the establishment of WWF.</p>
```

```
</section>
```

